

continuation

Continuation Document

Revision: 2 **Last modified:** 2026-05-31T00:25:00Z **Description:** Sacred invariant resumption document for Helix Cluster OS **Authority:** Constitution §12.10
Maintainer: Operator + AI loop

§0: How to use this document

Any CLI agent resuming work on this project **MUST** read this file first. It contains the canonical project state, recent history, active work, and known blockers. Do not assume state from external sources — this document is the single source of truth for session resumption.

To resume: 1. Read this file in full. 2. Check `docs/issues.md` for active workable items. 3. Check `docs/fixed.md` for recently completed work and resolution patterns. 4. Verify git state with `git status` and `git log --oneline -10`. 5. Proceed to the highest-priority active item from §3 below.

§1: Project State Summary

Property	Value
Project	Helix Cluster OS
Current Phase	Phase 3 (Session Manager) — extending into Phase 2 gaps
Overall Completion	~65% of MVP
Last Commit	9982c59 — MVP Phase 2-3: Scheduler, Resource Aggregator, Session Backends
Branch	main
Packages	33 Go packages, all tests pass with <code>-race</code>
Lines of Code	~14,040 in pkg/
Test Files	33 (100% package coverage)
Commits on main	16

§2: Recently Completed Work (last 10 commits)

Commit	Message	HXC
9982c59	MVP Phase 2-3: Scheduler, Resource Aggregator, Session Backends	HXC-204, HXC-205, HXC-206
f7dd99e	docs: MVP implementation progress tracker	HXC-901
35de8ef	Anti-bluff follow-up: Fix 8 stub packages + mutation tests for 15 packages	HXC-304, HXC-305
9aaeb9f	Anti-bluff audit follow-up: Real JWT implementation + mutation tests for 6 packages	HXC-303
e20c3ef	Constitution §7 + §11.4.102: Add inheritance verification gate + paired mutation test + anti-bluff audit	HXC-301, HXC-302
2e861d1	chore: cascade Helix Constitution inheritance pointers to all submodules	HXC-900
2632b1c	Phase 2: Core Infrastructure - SWIM protocol, WireGuard mesh, session manager, discovery race fixes, infra mutex	HXC-200, HXC-201, HXC-202, HXC-203
9bb47e2	Phase 3 Research.	HXC-300
8898518	Constitution: PCS-6 End-User Usability Guarantee	HXC-003
8185d0a	Phase 1: §11.4.29 snake_case compliance — rename kebab-case files and references	HXC-104

§3: Active Work

HXC	Title	Status	Priority	Details
HXC-019	Phase 7 QA & Testing	In progress	P0	Mutation test coverage expansion, integration test harness. All 33 packages pass <code>-race</code> . Need integration tests.
HXC-902	Documentation chain setup	In progress	P1	<code>docs_chain</code> infrastructure, CONTINUATION.md, Issues/Fixed tracking, HTML/PDF exports. Nearly complete.

HXC	Title	Status	Priority	Details
HXC-903	Continuation document maintenance	In progress	P1	Automated updates, revision headers, freshness checks. In progress.
HXC-016	Phase 4 Build Service	Queued	P0	Bazel RBE protocol, artifact caching, multi-arch builds. Next major deliverable.
HXC-017	Phase 5 LLM Brain	Queued	P1	Agent orchestration, inference scheduling. Deferred until Build Service complete.
HXC-018	Phase 6 Security Hardening	Queued	P1	mTLS everywhere, RBAC expansion, secret rotation. Can proceed in parallel.
HXC-020	Phase 8 Polish & Release	Queued	P2	v1.0.0-dev-mvp tag, release notes, final docs. Final milestone.

§4: Next Planned Work

1. **Commit current doc infrastructure** (HXC-902, HXC-903)
2. **Phase 4 Build Service** — Start Bazel RBE protocol implementation (HXC-016)
3. **Phase 6 Security** — mTLS config, certificate management (HXC-018)
4. **Integration tests** — Cross-package integration test suite (HXC-019)
5. **Phase 5 LLM Brain** — After Build Service reaches MVP (HXC-017)

§5: Environment Notes

- **Project root:** /Users/milosvasic/Projects/HelixCluster
- **Go workspace:** go.work with 25 modules
- **Test command:** go test ./pkg/... -race -count=1
- **Build command:** go build ./...
- **Commit wrapper:** bash scripts/commit_all.sh "message"
- **Doc generation:** bash scripts/docs/generate.sh
- **Doc verification:** bash scripts/docs/verify.sh
- **HXC tickets:** bash scripts/hxc.sh [next|list|show|create]
- **Docker:** Not available in build environment
- **golangci-lint:** Not available (skipped in pre-commit)
- **All GitHub Actions:** Disabled (manual CI/CD only)

§6: Known Issues / Blockers

- **None currently.** All 33 packages pass tests with race detector.
- **Docker unavailable** — Compose validation is structural only.

- **Third-party submodule pushes fail** — Expected (no write access to upstream repos).

§7: Quick Commands for Resumption

Verify state

```
cd /Users/milosvasic/Projects/HelixCluster
git status
git log --oneline -10
go test ./pkg/... -race -count=1
```

Check active work

```
bash scripts/hxc.sh list active
cat docs/issues.md
```

Generate docs

```
bash scripts/docs/generate.sh
bash scripts/docs/verify.sh
```

Constitution gate

```
bash tests/constitution/test_inheritance.sh
bash scripts/testing/meta_test_inheritance.sh
```